

Self-Evolving Constrained Distributed Optimization: Predictive Constraint Evolution under Dynamic Feasible Regions

Abstract—We propose SECDO, a predictive optimization framework for evolving constraints. SECDO (i) learns future constraint dynamics, (ii) performs predictability-conditioned anticipatory / mixed projection, and (iii) provides a dynamic regret upper bound with graceful degradation under prediction failures. Rather than claiming universal superiority, SECDO exhibits a favorable optimality–safety trade-off under evaluated dynamic-constraint scenarios and remains robust when forecasts become unreliable.

Index Terms—Online optimization, dynamic constraints, anticipatory projection, dynamic regret, learning-augmented optimization, UAV networks

I. INTRODUCTION

Constrained online optimization often treats the feasible set $\mathcal{B}_t$ as known, static, or revealed only after a decision. In cyber-physical systems such as multi-UAV bandwidth allocation under time-varying SINR, the binding constraint is itself dynamical: the scalar capacity $c_t = W \log_2(1 + \overline{\text{SINR}}_t)$ induces $\mathcal{B}_t = \mathcal{B}(c_t) = \{x \geq 0 : \mathbf{1}^\top x \leq c_t\}$. Reactive projection onto c_t lags set motion; brittle forecast-then-project pipelines can amplify violations when forecasts fail.

We study **Self-Evolving Constrained Distributed Optimization (SECDO)**: a *predictive constraint-evolution* framework for dynamic feasible optimization—not a prediction-enhanced projected-gradient module. SECDO identifies constraint evolution, anticipates $\hat{\mathcal{B}}_{t+1}$, and projects onto a predictability-conditioned mixed budget:

$$\mathcal{B}_t \rightarrow \hat{\mathcal{B}}_{t+1} \rightarrow \Pi \rightarrow x_{t+1}. \quad (1)$$

Let $\delta_t = d_H(\hat{\mathcal{B}}_{t+1}, \mathcal{B}_{t+1})$ denote feasible-set mismatch, $\chi_t = d_H(\mathcal{B}_{t+1}, \mathcal{B}_t)$ the constraint drift, and $\text{PI}_t = \delta_t / (\chi_t + \varepsilon)$ the predictability index. The weight $\alpha_t = 1 / (1 + \text{PI}_t^2)$ continuously interpolates anticipatory and reactive budgets.

Contributions. (i) We formulate evolving constraints as a predictive optimization problem and relate learned set mismatch to feasible-set uncertainty (Theorem 1). (ii) We develop SECDO with anticipatory / mixed projection and establish a dynamic regret upper bound governed by drift and set mismatch (Theorem 2). (iii) We characterize when anticipation tightens *feasibility certificates* via predictability ($\delta_t < \chi_t$) and provide failure-safe degradation (Theorem 3, Corollary 1). Empirically, a $13\times$ increase in mean drift yields a $56.7\times$ increase in cumulative dynamic regret with violation below 2%; crash tests and ablations isolate anticipation and failure containment.

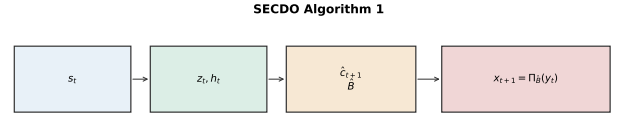


Fig. 1. SECDO loop: state encoding, constraint prediction, predictability-conditioned mixed projection.

II. PROBLEM FORMULATION

A. Static Objective, Dynamic Constraints

At each t ,

$$\min_{x \in \mathbb{R}^n} F_t(x) \quad \text{s.t.} \quad x \in \mathcal{B}_t = \mathcal{B}(c_t) = \{x \geq 0 : \mathbf{1}^\top x \leq c_t\}. \quad (2)$$

Let $x_t^* \in \arg \min_{x \in \mathcal{B}_t} F_t(x)$. We do *not* claim convergence to a moving unconstrained optimum.

B. UAV Capacity Teacher

$$c_t = W \log_2(1 + \overline{\text{SINR}}_t), \quad \chi_t = |c_{t+1} - c_t| = d_H(\mathcal{B}_{t+1}, \mathcal{B}_t). \quad (3)$$

C. Learned Constraint Dynamics

$$\hat{c}_{t+1} = \mathcal{F}_\phi(h_t), \quad \delta_t = d_H(\hat{\mathcal{B}}_{t+1}, \mathcal{B}_{t+1}) \leq |\hat{c}_{t+1} - c_{t+1}|. \quad (4)$$

SECDO establishes a controllable relationship between capacity residuals and feasible-set mismatch δ_t —it does *not* claim to learn the true future constraint exactly.

III. THEORETICAL ANALYSIS

We chain five objects: constraint identifiability $\rightarrow$ projection perturbation $\rightarrow$ dynamic regret $\rightarrow$ predictability-conditioned advantage $\rightarrow$ failure-safe recovery. Assumptions are collected in Appendix I (cf. Assumps. 1–4 in the supplement markdown freeze).

a) Notation freeze. $\delta_t := d_H(\hat{\mathcal{B}}_{t+1}, \mathcal{B}_{t+1})$ (feasible-set mismatch); $\chi_t := d_H(\mathcal{B}_{t+1}, \mathcal{B}_t)$ (constraint drift); $\epsilon_t := \|s_{t+1} - \hat{s}_{t+1}\|$ (state prediction error). Capacity residual $|\hat{c}_{t+1} - c_{t+1}|$ upper-bounds δ_t for scalar budgets (Theorem 1) and is *not* denoted δ_t .

A. Constraint Evolution Identifiability

Theorem 1 (Constraint evolution stability). *For scalar budgets $\mathcal{B}(c) = \{x \geq 0 : \mathbf{1}^\top x \leq c\}$,*

$$\delta_t = d_H(\mathcal{B}(c_{t+1}), \mathcal{B}(\hat{c}_{t+1})) \leq |c_{t+1} - \hat{c}_{t+1}|. \quad (5)$$

If $c = T(s)$ is L_g -Lipschitz and $\hat{c}_{t+1} = T(\hat{s}_{t+1})$, then $\delta_t \leq L_g \epsilon_t$.

Theorem 1 establishes a controllable relationship between capacity / state residuals and feasible-set mismatch δ_t , not that the predictor recovers the true future constraint. Proof: Appendix A.

Lemma 1 (Surrogate consistency). *Let $L_c = |\hat{c} - c|^2$ be the constraint regression loss and assume the scalar-budget geometry of Theorem 1. Then*

$$\mathbb{E}[\delta_t] \leq \mathbb{E}|\hat{c}_t - c_t| \leq \sqrt{\mathbb{E}[L_c]}. \quad (6)$$

Hence minimizing the constraint prediction loss contracts the theoretical set-mismatch term appearing in Theorems 1–2. This does not claim that minimizing L_c minimizes Reg_T .

Proof: Appendix B.

B. Projection Error Propagation

Lemma 2 (Projection perturbation). *Under bounded projection sensitivity, for any y ,*

$$\|\Pi_{\hat{\mathcal{B}}}(y) - \Pi_{\mathcal{B}}(y)\| \leq C_\Pi d_H(\hat{\mathcal{B}}, \mathcal{B}). \quad (7)$$

Consequently, with $x_{t+1} = \Pi_{\hat{\mathcal{B}}_{t+1}}(y_t)$ and $x_{t+1}^\circ = \Pi_{\mathcal{B}_{t+1}}(y_t)$, $\|x_{t+1} - x_{t+1}^\circ\| \leq C_\Pi \delta_t$. Under μ -strong convexity, $\|x_{t+1}^ - x_t^*\| \leq \mu^{-1}(\omega_t + L_g \chi_t)$.*

Remark 1. *The constant C_Π characterizes the sensitivity of the projection operator rather than a universal property of arbitrary convex sets. This assumption is not required for arbitrary convex sets, but holds for the polyhedral feasible regions considered in this work; boundedness then follows from standard Hoffman-type error bounds, and often $C_\Pi = O(1)$.*

Proof: Appendix C.

C. Dynamic Variational Regret

Define $\text{Reg}_T = \sum_{t=0}^{T-1} (F_t(x_t) - F_t(x_t^*))$ and $P_T = \sum_t \|x_{t+1}^* - x_t^*\|$.

Theorem 2 (Dynamic regret upper bound). *SECDO admits a dynamic regret upper bound governed by objective drift, constraint evolution, and feasible-set mismatch:*

$$\text{Reg}_T \leq O\left(\sqrt{T(1 + P_T)}\right) + O\left(\sum_t (\epsilon_t + \delta_t)\right), \quad (8)$$

with

$$P_T \leq \frac{1}{\mu} \sum_t (\omega_t + L_g \chi_t). \quad (9)$$

We do *not* claim that SECDO achieves optimal regret. Proof: Appendix D.

Algorithm 1 SECDO with Predictively Adaptive Projection (implementation uses a continuous relaxation of the theoretical binary regimes $\alpha \in \{0, 1\}$).

Require: step-size η ; predictor $\mathcal{F}_\phi$; encoder/GRU; objective F

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1: initialize  $x_0 \in \mathcal{B}(c_0)$ ,  $h_{-1} = 0$ 
2: for  $t = 0, 1, \dots, T - 1$  do
3:    $z_t \leftarrow \text{Enc}(s_t)$ ,  $h_t \leftarrow \text{GRU}(z_t, h_{t-1})$ 
4:    $\hat{s}_{t+1} \leftarrow \text{Dec}(h_t)$ ,  $\hat{c}_{t+1} \leftarrow \mathcal{F}_\phi(h_t)$ 
5:    $y_t \leftarrow x_t - \eta \nabla F(x_t)$ 
6:   estimate  $\hat{\delta}_t, \hat{\chi}_t$ ;  $\widehat{\text{PI}}_t \leftarrow \hat{\delta}_t / (\hat{\chi}_t + \varepsilon)$ 
7:    $\alpha_t \leftarrow 1 / (1 + \widehat{\text{PI}}_t^2)$ 
8:    $c_t^{\text{mix}} \leftarrow \alpha_t \hat{c}_{t+1} + (1 - \alpha_t) c_t$ 
9:    $x_{t+1} \leftarrow \Pi_{\mathcal{B}(c_t^{\text{mix}})}(y_t)$ 
10:  observe  $s_{t+1}, c_{t+1}$ ; optional online update of  $\phi$  with  $\|\Delta\theta\| \leq \zeta$ 
11: end for
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D. Predictability-Conditioned Feasibility Improvement

Let $V_R(t) \leq \chi_t$ and $V_A(t) \leq \delta_t$ be reactive / anticipatory future-violation certificates, and $\text{PI}_t = \delta_t / (\chi_t + \varepsilon)$.

Theorem 3 (Predictability-conditioned feasibility improvement). *If $\text{PI}_t < 1$ (i.e., $\delta_t < \chi_t$), anticipatory projection provides a tighter constraint-violation certificate when set mismatch is smaller than constraint drift:*

$$\delta_t < \chi_t \Rightarrow \text{anticipatory feasibility certificate strictly tighter than reactive.} \quad (10)$$

This is a statement about feasibility certificates, not about objective superiority, and not a claim that SECDO is always better. Proof: Appendix E.

E. Failure-Safe Recovery

Corollary 1 (Graceful degradation). *With $\alpha_t = 1 / (1 + \text{PI}_t^2)$ and $c_t^{\text{mix}} = \alpha_t \hat{c}_{t+1} + (1 - \alpha_t) c_t$, when $\text{PI}_t \gg 1$ one has $\alpha_t \rightarrow 0$ and SECDO gracefully degrades toward reactive behavior. Writing $I_t^{\text{fail}} = \mathbf{1}\{\text{PI}_t > 1\}$,*

$$\text{Reg}_{\text{SECDO}} \leq \text{Reg}_{\text{Reactive}} + O\left(\sum_t I_t^{\text{fail}} \delta_t\right). \quad (11)$$

We do not claim full recovery to a failure-free trajectory. Proof: Appendix F.

IV. ALGORITHM

Remark 2 (Theory vs. implementation). *The ideal anticipatory analysis in Theorems 2–3 compares projections onto $\hat{\mathcal{B}}_{t+1}$ and $\mathcal{B}_t$ (extremes corresponding to $\alpha = 1$ and $\alpha = 0$). The deployed Algorithm 1 uses budget mixing c^{mix} with $0 < \alpha_t \leq 1$, which continuously interpolates those extremes and enables Corollary 1. The mixed feasible region preserves a single projection operation, avoiding additional optimization overhead from combining two projected points. Adaptive α is an implementation-level robustness mechanism; the core certificates characterize ideal anticipatory vs. reactive projection.*

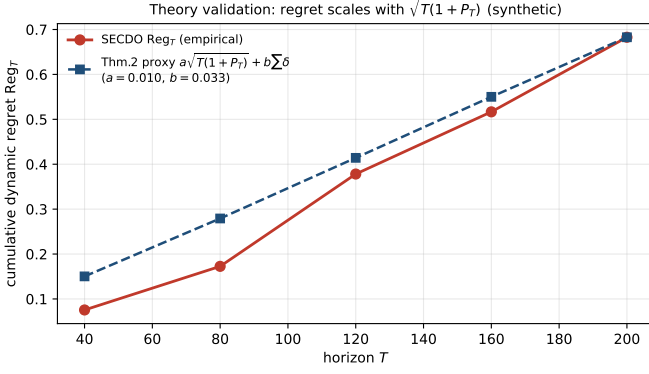


Fig. 2. Dynamic regret scaling under synthetic evolving constraints. Empirical Reg_T tracks the structure $O\sqrt{T(1+P_T)} + O\sum\delta$. The fitted envelope is only used for visualization and does not represent a universal constant.

TABLE I
SECDO UNDER INCREASING ENVIRONMENTAL DRIFT (5 SEEDS).

Regime	$\bar{\chi}$	Reg_T	Violation
Slow	0.010	7.69×10^{-4}	0.33%
Fast	0.131 ($\times 13.0$)	4.36×10^{-2} ($\times 56.7$)	1.46%
Stress	0.203	6.86×10^{-2}	1.89%

Training $\neq$ online algorithm. Offline, we minimize $L = \lambda_s L_s + \lambda_c L_c + \lambda_u L_u$ with $(\lambda_s, \lambda_c, \lambda_u) = (0.5, 1.0, 0.1)$. Minimizing L_c contracts δ via Lemma 1 but does not directly minimize Reg_T . During training, $\hat{c}$ is stop-grad through Π . Details: Appendix I.

V. EXPERIMENTS

SECDO does not target Pareto dominance on all metrics. We evaluate (i) scaling structure on synthetic convex problems, (ii) optimality–safety trade-off on UAV allocation, and (iii) prediction-failure stress tests.

A. Experimental Setup

Synthetic convex: theory validation of Theorem 2 scaling and PI behavior—not leaderboard tuning. **UAV system evaluation:** practical benefit and safety/efficiency trade-off under Slow/Fast/Stress regimes (5 seeds); this demonstrates system behavior only (synthetic experiments address scaling laws). **Prediction failure stress test:** Corollary 1 via corrupted $\hat{c}$ windows.

B. Dynamic Regret Scaling

Fig. 2 demonstrates the *scaling behavior* predicted by Theorem 2. Coefficients in the plotted envelope are fitted only for visualization.

C. Drift Sensitivity on UAV

Table I: a $13\times$ rise in $\bar{\chi}$ induces a $56.7\times$ rise in Reg_T with bounded violation.

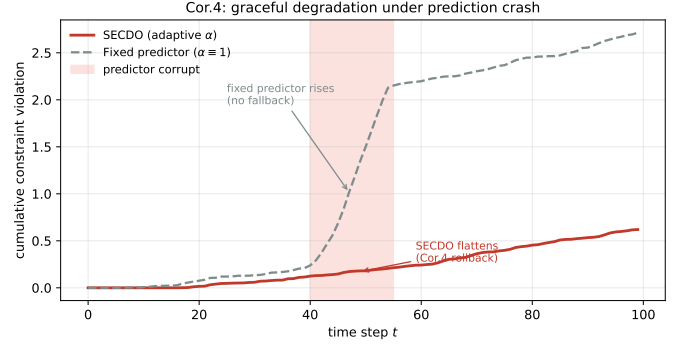


Fig. 3. Robust recovery under predictor failure (crash protocol). SECDO gracefully degrades relative to fixed $\alpha \equiv 1$.

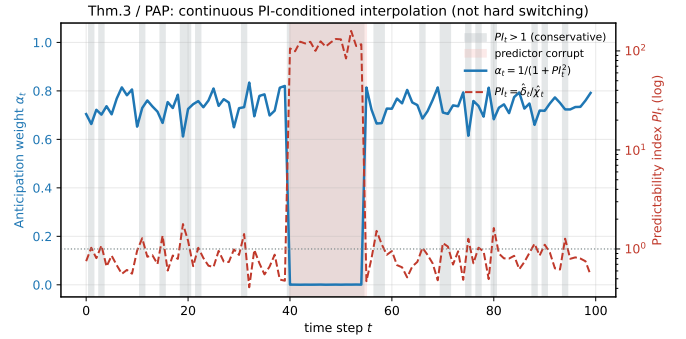


Fig. 4. Online PI_t and adaptive weight $\alpha_t = 1/(1 + \text{PI}_t^2)$ under predictor corruption.

D. Failure Recovery and PI Adaptation

Figs. 3–4 demonstrate the empirical behavior predicted by Corollary 1 and predictability-conditioned mixing.

E. Optimality–Safety Trade-off

Fig. 5: SECDO reduces optimality gap by exploiting constraint evolution, at a moderate violation cost versus reactive methods—the expected anticipatory vs. conservative trade-off.

F. Ablation

Table II: removing anticipation increases Fast regret by $\sim 18\%$. Adaptive α mainly improves robustness rather than nominal performance: when predictions are accurate, $\alpha \equiv 1$ is nearly unchanged; under failure it prevents error amplification (Corollary 1).

VI. CONCLUSION

SECDO is a predictive constraint-evolution framework for dynamic feasible optimization: it learns $\hat{B}_{t+1}$, applies predictability-conditioned mixed projection, and admits a dynamic regret guarantee with graceful degradation under prediction failure. Theory (Theorems 1–3, Corollary 1), Algorithm 1, and experiments form a single chain—constraint evolution, anticipatory / mixed projection, adaptive robustness, and regret analysis—rather than a stack of unrelated heuristics.

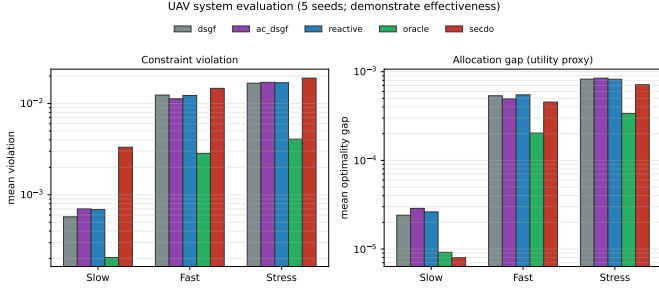


Fig. 5. Optimality–safety trade-off under dynamic constraints (UAV, 5 seeds). SECDO achieves a favorable gap–violation trade-off relative to conservative reactive baselines. Oracle uses future c_{t+1} (informational upper bound).

TABLE II
ABLATION ON FAST UAV REGRET AND CRASH-WINDOW
CUMULATIVE-VIOLATION RISE.

Variant	Fast Reg_T	Crash window rise
Full SECDO	0.044	0.185
A1/A2 (no anticipation)	0.052 (+18%)	0.142
A3 ($\alpha \equiv 1$)	0.043	0.559 ($\times 3$)

a) *Limitations.*: SECDO currently assumes that evolving feasible regions admit bounded projection sensitivity and focuses on online optimization with predictable constraint evolution under polyhedral (scalar-budget) geometry. Extending the framework to highly nonconvex feasible manifolds, stronger partial observability, and fully coupled classical topology solvers remains an interesting direction. UAV baselines for DSGF/AC-DSGF are evaluated as myopic adapters under a shared capacity teacher and should not be read as exhaustive reproductions of every historical variant.

APPENDIX A PROOF OF THEOREM 1

Assume without loss of generality $c \geq \hat{c} \geq 0$. Then $\mathcal{B}(\hat{c}) \subseteq \mathcal{B}(c)$, so $\sup_{z \in \mathcal{B}(\hat{c})} \text{dist}(z, \mathcal{B}(c)) = 0$. For $x \in \mathcal{B}(c)$ with $\mathbf{1}^\top x > \hat{c}$, the radial map $\tilde{x} = (\hat{c}/\mathbf{1}^\top x)x$ lies in $\mathcal{B}(\hat{c})$ and $\|x - \tilde{x}\|_2 \leq c - \hat{c}$. Hence $d_H(\mathcal{B}(c), \mathcal{B}(\hat{c})) \leq |c - \hat{c}|$. The Lipschitz claim follows from Assump. 1: $|T(\hat{s}) - T(s)| \leq L_g \|\hat{s} - s\|$.

APPENDIX B PROOF OF LEMMA 1

By Theorem 1, $\delta_t \leq |\hat{c}_t - c_t|$ under scalar budgets. Taking expectations and applying Jensen’s inequality / Cauchy–Schwarz on the scalar residual yields

$$\mathbb{E}[\delta_t] \leq \mathbb{E}|\hat{c}_t - c_t| \leq \sqrt{\mathbb{E}|\hat{c}_t - c_t|^2} = \sqrt{\mathbb{E}[L_c]}. \quad (12)$$

Thus reducing L_c contracts the set-mismatch term that enters Theorems 1–2. We emphasize that L_c is a surrogate for δ , not a surrogate for Reg_T .

APPENDIX C PROOF OF LEMMA 2

By Assump. 3 (bounded projection sensitivity), for fixed y , $\|\Pi_{\hat{\mathcal{B}}}(y) - \Pi_{\mathcal{B}}(y)\| \leq C_{\Pi} d_H(\hat{\mathcal{B}}, \mathcal{B})$. Applying this at $y = y_t$

with $\hat{\mathcal{B}} = \hat{\mathcal{B}}_{t+1}$ and $\mathcal{B} = \mathcal{B}_{t+1}$ yields $\|x_{t+1} - x_{t+1}^o\| \leq C_{\Pi} \delta_t$. The comparator motion bound follows from μ -strong convexity of F_t and Assump. 4 (bounded ω_t, χ_t), as in standard gradual-variation arguments for strongly convex problems. As noted in Remark 1, Assump. 3 is not claimed for arbitrary convex sets; it is used here for the polyhedral (scalar-budget) regions studied in this paper.

APPENDIX D PROOF SKETCH OF THEOREM 2

Start from a projected-gradient inequality for the *exact* projection onto $\mathcal{B}_{t+1}$. Replace the exact projection by SECDO’s projection onto $\hat{\mathcal{B}}_{t+1}$ (or $\mathcal{B}(c^{\text{mix}})$) using Lemma 2; Lipschitz continuity of F_t contributes an additive $O(\delta_t)$ term per step. State-prediction noise in $\hat{\nabla}_t$ contributes $O(\epsilon_t)$. Telescope distance terms and apply the standard dynamic-regret / path-variation argument with $P_T = \sum \|x_{t+1}^* - x_t^*\|$, choosing $\eta \asymp \sqrt{(1 + P_T)/T}$ (or an equivalent two-stage bound), to obtain $O\sqrt{T(1 + P_T)} + O\sum(\epsilon_t + \delta_t)$. The bound on P_T is Lemma 2. Full expansions mirror the SECDO theory freeze notes (thm2_prediction_aware_gap.md).

APPENDIX E PROOF OF THEOREM 3

For reactive $x_{t+1}^R = \Pi_{\mathcal{B}_t}(y_t)$, $\text{dist}(x_{t+1}^R, \mathcal{B}_{t+1}) \leq d_H(\mathcal{B}_t, \mathcal{B}_{t+1}) = \chi_t$, so $V_R(t) \leq \chi_t$. For anticipatory $x_{t+1}^A = \Pi_{\hat{\mathcal{B}}_{t+1}}(y_t) \in \hat{\mathcal{B}}_{t+1}$, $\text{dist}(x_{t+1}^A, \mathcal{B}_{t+1}) \leq \delta_t$, so $V_A(t) \leq \delta_t$. If $\delta_t < \chi_t$, the anticipatory *feasibility* upper-bound certificate is strictly tighter. This compares constraint-violation certificates, not objective superiority, and not pathwise distances on every sample.

APPENDIX F PROOF SKETCH OF COROLLARY 1

With $c_t^{\text{mix}} = \alpha_t \hat{c}_{t+1} + (1 - \alpha_t)c_t$ and $\alpha_t = 1/(1 + \text{PI}_t^2)$,

$$|c_t^{\text{mix}} - c_{t+1}| \leq \alpha_t \delta_t^{\text{cap}} + (1 - \alpha_t)\chi_t. \quad (13)$$

As $\text{PI}_t \rightarrow \infty$, $\alpha_t \rightarrow 0$ and the update degrades to reactive projection onto $\mathcal{B}_t$. Comparing per-step regret to a pure reactive baseline, excess cost on failure indicators I_t^{fail} is absorbed as $O(I_t^{\text{fail}} \delta_t)$, yielding the stated cumulative bound. SECDO *gracefully degrades* toward reactive behavior; it does not claim full recovery.

APPENDIX G ADDITIONAL EXPERIMENTS

Fig. 6 expands Table II. A1/A2 coincide operationally when $\alpha = 0$ (projection onto $\mathcal{B}(c_t)$). A3 ($\alpha \equiv 1$) nearly matches Full on clean Fast UAV but inflates crash-window rise by $\approx 3\times$, isolating failure containment.

a) *Evidence chain.*: Claims map to theory and figures as: Thm. 1 $\leftrightarrow \delta$ logs; Lem. 2 $\leftrightarrow$ Oracle gap; Thm. 2 $\leftrightarrow$ Fig. 2 & Table I; Thm. 3 $\leftrightarrow$ Fig. 4; Cor. 1 $\leftrightarrow$ Fig. 3 & A3.

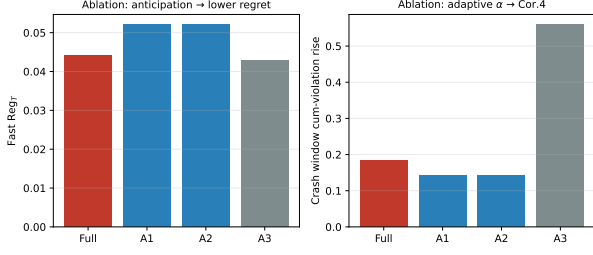


Fig. 6. Ablation summary: Fast-regime regret (left) and crash-window cumulative-violation rise (right). Demonstrates the empirical behavior predicted by Corollary 1 (adaptive α vs. $\alpha \equiv 1$).

APPENDIX H COMPUTATIONAL COMPLEXITY

Let $O(P)$ denote one Euclidean projection onto a scalar-budget simplex / polytope (sorting-based, $O(n \log n)$ typical), and $O(d)$ a cheap first-order update (gradient step in allocation dimension n). Let $O(C_\phi)$ be one forward pass of the constraint predictor $\mathcal{F}_\phi$.

TABLE III
PER-ITERATION COMPLEXITY (SCHEMATIC).

Method	Prediction	Projection	Update
Reactive	—	$O(P)$	$O(d)$
Oracle	future c_{t+1} known	$O(P)$	$O(d)$
SECDO	$O(C_\phi)$ inference	$O(P)$ (single)	$O(d)$

Relative to reactive projection, SECDO adds predictor inference $O(C_\phi)$ and scalar PI/α arithmetic. It does *not* increase the number of projections: the mixed budget c^{mix} induces one set $\mathcal{B}(c^{\text{mix}})$ and a single Π . Thus prediction does not make the optimization subproblem asymptotically more expensive than reactive / oracle projection under the same geometry.

APPENDIX I IMPLEMENTATION CONSISTENCY

a) *Assumptions (summary)*.: Assump. 1: Lipschitz constraint/teacher map (L_g). Assump. 2: μ -strong convexity and L_F -smoothness of F_t . Assump. 3: bounded projection sensitivity C_Π (Remark 1). Assump. 4: bounded drifts χ_t, ω_t .

b) *Code map*.: $\hat{c}$: capacity head; $\text{PI}, \alpha, c^{\text{mix}}$: `secdo.optimizer.secdo_optimizer`; Π : scalar-budget projection; L_c : constraint regression loss. Oracle alone peaks c_{t+1} ; SECDO never does. Version tag: SECDO-v2.0; environment: `conda pytorch12, CUDA`.

c) *Simplified logical chain*.:

learned constraint evolution →
anticipatory / mixed projection →
adaptive robustness → regret analysis.

We intentionally avoid presenting PI , α , recovery, etc. as independent stacked heuristics.